

Atomic Evaluation using Large Language Model for Automated Essay Exam Scoring

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Abstract—Education is a cornerstone of national development and plays a crucial role in achieving sustainable development goals. Traditional manual grading of exams and assignments, particularly for subjective responses like essays, is time-consuming, resource-intensive, and prone to human biases and inconsistencies. To address these challenges, Automated Essay Scoring (AES) systems have been developed, leveraging advancements in machine learning, deep learning, and particularly in Large Language Models (LLMs). However, small parameter LLMs often struggle to maintain coherence over longer texts and accurately capture nuanced details, resulting in lower agreement rates with human raters. This study proposes a novel approach called Atomic Evaluation to enhance the performance of small parameter LLMs in AES tasks. Atomic Evaluation decomposes student responses into smaller, more manageable atomic units, enabling more granular and focused evaluations. By aligning these atomic units with detailed rubrics and using a multi-step scoring process, the method improves the precision and transparency of automated scoring. Experiments conducted on two datasets, eLOK and ASAP-SAS, using Llama 3 8B and Gemma 2 9B models, show that Atomic Evaluation significantly improves assessment accuracy, with an increase in Quadratic Weighted Kappa (QWK) values of up to 0.08 for Llama 3 8B and 0.23 for Gemma 2 9B. These results demonstrate the potential of Atomic Evaluation to provide a more reliable and efficient AES solution, especially in educational environments with limited computing resources.

Index Terms—Atomic Evaluation, Automated Essay Exam Evaluation, Large Language Model, Prompt Technique

I. INTRODUCTION

Education is fundamental to national development and achieving sustainable development goals. With the increasing integration of digital technology into education—such as online courses and e-learning platforms—there is a growing need for efficient, scalable, and consistent assessment methods. Traditional manual grading especially for subjective responses like essays is time-consuming, resource-intensive, and prone to human biases and inconsistencies, potentially compromising the quality and fairness of education [1].

To address these challenges, Automated Essay Scoring (AES) systems have been developed, leveraging advance-

ments in Natural Language Processing (NLP) and Machine Learning. Since the inception of the Project Essay Grader (PEG) by Ellis Batten Page in 1966 [2], which pioneered computer-based essay evaluation, AES research has significantly evolved. Early systems like PEG used statistical models to identify text patterns and assigned scores based on features such as word count, syntax, and grammar. The field has advanced by incorporating Deep Learning and Large Language Models (LLMs), greatly enhancing the accuracy and contextual understanding of automated scoring.

Despite these advancements, small parameter LLMs still face significant limitations. While large parameter LLMs have demonstrated a high agreement rate with human raters, indicating better alignment in scoring, small parameter LLMs often produce suboptimal results. They struggle to maintain coherence over longer texts, capture nuanced details, and effectively follow instructions which is crucial in diverse real-world scenarios [3]. This discrepancy poses a challenge for developing a cost-effective and efficient automated scoring system suitable for environments with limited computational resources.

This research proposes a novel approach called Atomic Evaluation to address the limitations of small parameter LLMs. The primary objective is to enhance their performance by decomposing student responses into smaller, manageable atomic units for focused and accurate evaluation. This method aims to improve the agreement rate between the model and human raters, making small parameter LLMs more effective in educational settings where computational resources are constrained.

II. RELATED WORKS

Over the years, Automated Essay Scoring (AES) has evolved from using machine learning methods to employing deep learning techniques and, more recently, advanced Large Language Models (LLMs).

A. Machine Learning

Early AES systems used interpretable machine learning models with shallow text features, focusing on statistical and

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grammatical aspects for essay scoring. Catulay et al. [4] combined neural networks with Bayesian linear ridge regression to enhance AES systems' correctness and reliability, using specific features and a Hybrid Agile model for development. Thamrin et al. [5] used classification techniques and similarity algorithms for AES in Bahasa Indonesia, assessing their effectiveness via RMSE values, highlighting AES application in multilingual contexts. Goura et al. [6] used Multiple Linear Regression and XGBoost to develop an AES system evaluating essays based on rubrics like cohesion, coherence, and idea development, offering comparative analysis of recent models. Li et al. [7] proposed a stacking method-based AES model combining multiple models to outperform traditional models in accuracy. While these machine learning methods are interpretable, they can be limited with complex, open-ended essay topics.

B. Deep Learning

Deep learning advances AES by enabling models to learn complex patterns and deep text features without extensive feature engineering. These models capture high-dimensional text embeddings, offering better scoring performance for diverse essay topics despite lacking direct linguistic interpretability. Li et al. [8] proposed a novel two-part model (SEDNN) for cross-prompt AES, effectively transferring rating knowledge between prompts and incorporating prompt-dependent features for more accurate scoring. Arifin et al. [9] showed that Siamese Manhattan LSTM effectively grades Indonesian short answers, achieving high accuracy and Quadratic Weighted Kappa scores, especially with limited data. Suresh et al. [10] combined LSTM networks with Random Forest and Deep Neural Networks, creating integrated frameworks that outperform single-model approaches in accuracy and consistency. Prabhu et al. [11] used a fine-tuned BERT model for AES, achieving better semantic understanding and improved scoring reliability.

Tambe et al. [12] introduced an AES system incorporating grammar error detection and structural evaluation using BERT embeddings, advancing AES by adding a grammatical analysis dimension. Xue et al. [13] combined a pre-trained BERT model with a capsule network for essay scoring, achieving state-of-the-art accuracy and robustness. Despite improved accuracy, these deep learning models require substantial computational resources and are less interpretable than traditional models. Bai et al. [14] presented an incremental learning framework integrating crowdsourcing for data collection to train AES models, addressing catastrophic forgetting and concept drift, offering a scalable solution. Mukti et al. [15] applied contrastive learning with BERT-based models for Indonesian short-answer grading, significantly improving accuracy by capturing semantic similarities.

C. Large Language Model

Large Language Models (LLMs) bring new capabilities to AES by understanding complex language patterns and providing detailed feedback, thanks to pre-training on vast

TABLE I
DATASET OVERVIEW

Name	Question	Answer	Rater	QWK
eLOK	15	2763	5	0.8025
ASAP-SAS	10	17207	2	0.9125

text data. Mizumoto et al. [16] showed the effective use of AI language models like GPT for AES, emphasizing LLMs' potential to enhance accuracy with linguistic features. Song et al. [17] presented prompt-tuning methods like Zero Shot, Few Shot, and P-Tuning V2, allowing training LLMs without altering parameters, showcasing their practical application in AES using open-source LLMs achieving competitive performance.

Additionally, the FACTSCORE method introduced by Min et al. [18] demonstrates a fine-grained atomic approach for evaluating factual precision in long-form text generation by breaking down content into atomic facts. While FACTSCORE focuses on evaluating the factual accuracy of model-generated content by verifying each atomic fact against external knowledge sources, the Atomic Evaluation approach in this study focuses on decomposing both questions and student responses into atomic units to assess correctness, completeness, and alignment with detailed rubrics. Both methods emphasize fine-grained evaluation, but FACTSCORE is primarily concerned with factual correctness, whereas Atomic Evaluation is designed to enhance scoring result and educational assessment in Automated Essay Scoring (AES) tasks. Adopting LLMs in AES offers better performance and flexibility but requires careful consideration of computational costs and scalability in educational settings.

III. MATERIAL AND METHOD

A. Dataset

As shown in Table I, two datasets were used to evaluate the Atomic Evaluation method: eLOK and ASAP-SAS. The eLOK dataset, a collection from a Learning Management System (LMS) named eLOK, which is used by Gadjah Mada University, includes mid-term and final exam data from the Digital Transformation course. It consists of 15 questions (6 from mid-terms and 9 from finals) and 2763 student responses, each scored by five human raters, yielding an average Quadratic Weighted Kappa (QWK) score of 0.8025. Its varied question types and real-world context make it ideal for testing model accuracy. In contrast, the ASAP-SAS dataset is a publicly available benchmark for short-answer scoring, containing 10 questions and 17,207 responses, with a higher QWK score of 0.9125. These datasets provide a robust foundation for evaluating model performance in both real-world and standardized settings.

1) *eLOK Dataset*: The eLOK dataset¹ is a dataset sourced from a Learning Management System (LMS) named eLOK, which is used by Gadjah Mada University. The dataset

comprises mid-term and final exam data from the Digital Transformation course, with a total of 15 questions—6 from mid-terms and 9 from final exams—and 2763 student responses. Each response is accompanied by a detailed scoring rubric, and five human raters scored the responses, resulting in an average Quadratic Weighted Kappa (QWK) score of 0.8025.

2) *ASAP-SAS Dataset*: The ASAP-SAS (Automated Student Assessment Prize - Short Answer Scoring) dataset is a publicly available dataset commonly used for short answer scoring tasks. It consists of 10 questions with a total of 17207 student responses. The dataset is annotated by two human raters, and their evaluations have a high inter-rater agreement with an average QWK score of 0.9125.

B. LLMs Selection

As shown in Table II, two Large Language Models (LLMs), Llama 3 8B by Meta AI and Gemma 2 9B by Google Research, were selected for this study based on their suitability for natural language processing tasks. Llama 3 8B, with 8 billion parameters, offers a balance between efficiency and accuracy, utilizing techniques like Supervised Fine-Tuning (SFT) and Reinforcement Learning with Human Feedback (RLHF) to enhance its alignment with human judgment. Gemma 2 9B, with 9 billion parameters, is optimized for speed and performance, using knowledge distillation and Grouped-Query Attention (GQA) to mimic larger models while maintaining computational efficiency. Both models were chosen for their ability to handle Automated Essay Scoring (AES) tasks efficiently in resource-constrained environments.

- 1) Llama 3 8B Developed by Meta AI, Llama 3 is part of a family of open-source models optimized for diverse text-generation tasks. The 8 billion parameter variant of Llama 3 was chosen for its balance between computational efficiency and performance. This model leverages techniques such as supervised fine-tuning (SFT) and reinforcement learning with human feedback (RLHF), which enhance its usability and alignment with human preferences [19].
- 2) Gemma 2 9B is a lightweight, state-of-the-art LLM developed by Google DeepMind, designed specifically to deliver high performance despite its relatively smaller size. The 9 billion parameter version of the model was selected for its efficiency in handling a wide range of language tasks. Key architectural innovations include the use of knowledge distillation, which allows the model to inherit the performance of much larger

models by mimicking their behavior, and Grouped-Query Attention (GQA), which increases inference speed without sacrificing performance [20].

C. Atomic Evaluation

The proposed Atomic Evaluation method aims to improve the performance of small-parameter LLMs by breaking down both questions and answers into smaller, atomic units for more focused and granular evaluation. Fig. 1 depicts the overall workflow of the Atomic Evaluation process, which consists of two main phases: Preprocessing and Evaluation.

1) *Preprocessing*: The preprocessing phase prepares the data for evaluation by transforming it into a format suitable for the model and ensuring that relevant learning materials are readily accessible for reference during the evaluation. This phase involves the following steps:

- ETL (Extract, Transform, Load): This step involves extracting relevant information such as questions, student answers, and rubrics from the dataset, transforming it into a format suitable for further processing, and loading it into the system. The ETL process ensures that only the necessary content is retained, improving the efficiency and effectiveness of subsequent evaluation steps.
- Perform Retrieval-Augmented Generation (RAG) to Finding the Ground Truth: Retrieval-Augmented Generation (RAG) is a hybrid model that combines parametric and non-parametric memory to enhance language generation for knowledge-intensive tasks [21].

As depicted in Fig. 2, learning materials from knowledge sources are embedded into vector representations and stored in a vector database. When a query is submitted by the application, it is passed to the search mechanism, which retrieves relevant information from the vector database. This retrieved information is used to enhance

TABLE II
SELECTED LLMs OVERVIEW

Name	Developer	Release Date
Llama 3 8b	Meta AI	18 April 2024
Gemma 2 9b	Google Research	27 June 2024

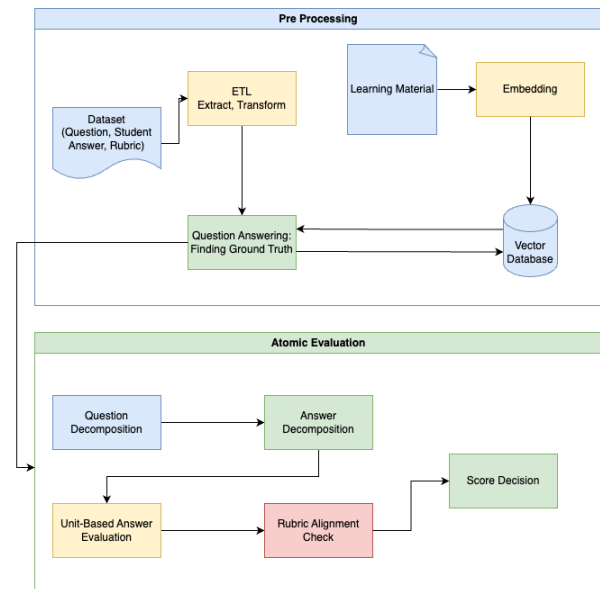


Fig. 1. Overview of the Atomic Evaluation.

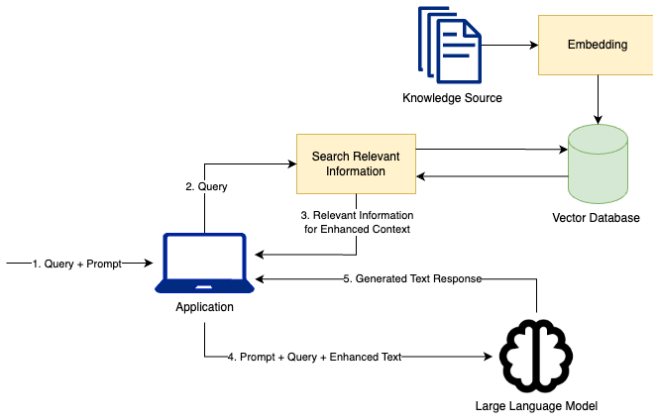


Fig. 2. Retrieval-Augmented Generation Overview.

the query's context before being sent to the Large Language Model (LLM) for generating a response. RAG helps prevent model being hallucination by grounding its generated outputs in these retrieved reference materials. This process ensures that the automated essay scoring system remains aligned with factual content.

2) *Evaluation*: The evaluation phase involves breaking down both questions and answers into smaller, atomic units, evaluating them independently, and then aggregating these evaluations to provide a comprehensive and accurate assessment. The steps involved in this phase are as follows:

- **Question Decomposition**: As shown in Fig. 3, the questions are split into individual sub-questions, each representing a distinct aspect or requirement. This step allows the model to focus on specific components of the question, improving the accuracy of the assessment by ensuring that each part is evaluated separately.
- **Student Answer Decomposition**: As shown in Fig. 4, the student's answer is divided into segments corresponding to the split questions. This alignment ensures that each segment of the student's response is directly evaluated

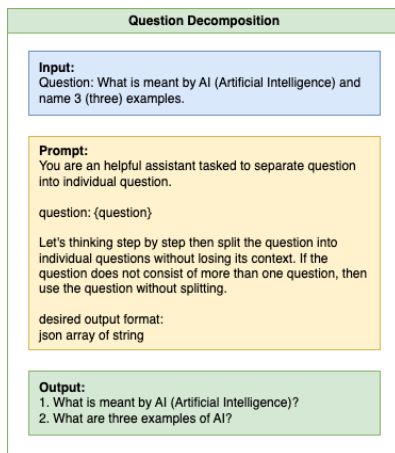


Fig. 3. Question Decomposition.

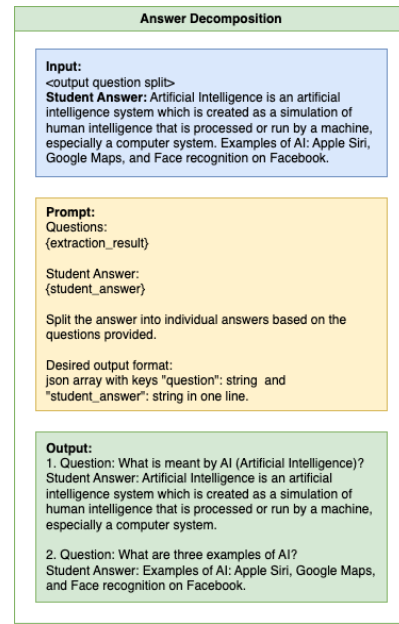


Fig. 4. Student Answer Decomposition.

against the relevant sub-question, providing a more focused and context-specific evaluation.

- **Unit-Based Answer Evaluation**: Each atomic unit of the student's answer is evaluated independently using a predefined scoring rubric and the retrieved ground truth. The model assigns a "correct rate" to each answer segment, indicating its accuracy, and provides an explanation to support the evaluation. This step ensures transparency in the scoring process by highlighting the reasoning behind each score.
- **Rubric Alignment Check**: The evaluated answers are then matched against a detailed rubric to determine whether they meet the expected standards. This process involves checking if each criterion in the rubric is met and calculating a "met rate." The rubric matching process ensures consistency and fairness in the evaluation by aligning the model's scoring with human-defined criteria.
- **Score Decision**: The final score for the student's response is determined by aggregating the results from the rubric matching process. The model selects "is met" with the highest score as the representative score and provides a comprehensive evaluation based on the best-matched rubric. This step provides a holistic assessment of the student's answer, integrating all evaluated components to form a final score.

D. Evaluation Metrics

This research result is evaluated using Quadratic Weighted Kappa (QWK), a metric commonly employed in Automated Essay Scoring (AES) to measure the agreement between model predictions and human raters. QWK is designed for ordinal rating tasks, where it adjusts for chance agreement

TABLE III
INTERPRETATION OF KAPPA

Kappa	Interpretation
< 0	Less than chance agreement
0.01 - 0.20	Slight agreement
0.21 - 0.40	Fair agreement
0.41 - 0.60	Moderate agreement
0.61 - 0.80	Substantial agreement
0.81 - 1.00	Almost perfect agreement

and penalizes disagreements based on their severity [22]. The QWK formula is expressed as:

$$QWK = 1 - \frac{\sum_{i,j} W_{i,j} O_{i,j}}{\sum_{i,j} W_{i,j} E_{i,j}} \quad (1)$$

where $W_{i,j}$ is the weight matrix based on the squared differences between categories i and j , $O_{i,j}$ is the observed agreement, and $E_{i,j}$ is the expected agreement by chance. This weighted approach provides a more nuanced assessment of performance compared to unweighted metrics.

Table III presents the interpretation of Kappa values, which are used to assess the level of agreement between the model and human raters. Kappa values range from less than 0, indicating less than chance agreement, to values between 0.81 and 1.00, which represent almost perfect agreement.

IV. RESULT AND DISCUSSION

A. Result on eLOK Dataset

Table IV compares the performance of the models using the CoT method and the proposed Atomic Evaluation method based on Quadratic Weighted Kappa (QWK) scores on eLOK dataset.

As shown in Fig. 5, The Llama 3 8B and Gemma 2 9B model demonstrates consistent improvement across all raters. The Llama 3 8B model demonstrates consistent improvement across all raters when applying the Atomic Evaluation method compared to the CoT method. The Quadratic Weighted Kappa (QWK) scores increased for all five raters, with an average QWK improvement of 0.0548.

The Gemma 2 9B model shows even more substantial improvements with the Atomic Evaluation method compared to the CoT method. The QWK scores increased significantly

TABLE IV
RESULT COMPARISON OF CoT AND ATOMIC EVALUATION METHODS ON eLOK DATASET

Model	Rater	CoT	Atomic Eval	Improvement
Llama 3 8B	Rater 1	0.40056	0.43085	+0.03029
	Rater 2	0.49430	0.52946	+0.03516
	Rater 3	0.37751	0.45819	+0.08068
	Rater 4	0.42445	0.49131	+0.06686
	Rater 5	0.42403	0.48482	+0.06079
Gemma 2 9B	Rater 1	0.44909	0.66377	+0.21468
	Rater 2	0.58540	0.76022	+0.17482
	Rater 3	0.45602	0.68510	+0.22908
	Rater 4	0.49461	0.69000	+0.19539
	Rater 5	0.50266	0.69307	+0.19041

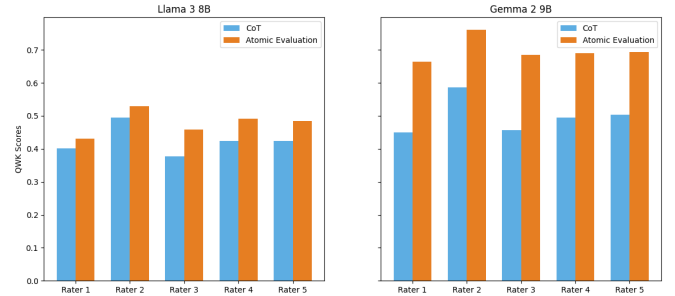


Fig. 5. Result Comparison of CoT and Atomic Evaluation Methods on eLOK Dataset.

across all five raters, with an average QWK improvement of 0.2009.

B. Result on ASAP-SAS Dataset

Table V compares the performance of the models using the CoT method and the proposed Atomic Evaluation method based on Quadratic Weighted Kappa (QWK) scores on ASAP-SAS dataset.

As shown in Fig. 6, The Llama 3 8B and Gemma 2 9B model demonstrates consistent improvement across all raters. The Llama 3 8B model demonstrates noticeable improvement when applying the Atomic Evaluation method compared to the CoT method. The Quadratic Weighted Kappa (QWK) scores increased for both raters, with an average QWK improvement of 0.0437.

The Gemma 2 9B model shows similar trends of improvement with the Atomic Evaluation method compared to the CoT method. The QWK scores also increased for both raters, with an average QWK improvement of 0.0299.

TABLE V
PERFORMANCE COMPARISON OF CoT AND ATOMIC EVALUATION METHODS ON ASAP-SAS DATASET

Model	Rater	CoT	Atomic Eval	Improvement
Llama 3 8B	Rater 1	0.34186	0.38047	+0.03861
	Rater 2	0.34637	0.39523	+0.04886
Gemma 2 9B	Rater 1	0.36583	0.39558	+0.02975
	Rater 2	0.37448	0.40445	+0.02997

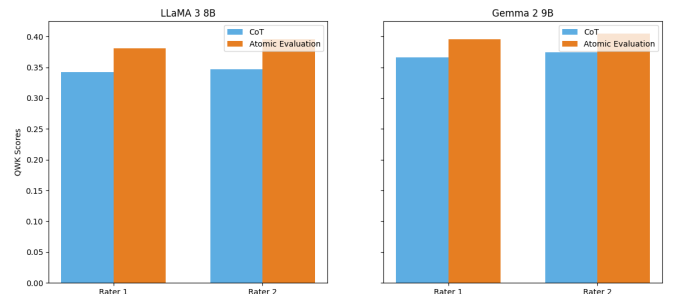


Fig. 6. Result Comparison of CoT and Atomic Evaluation Methods on eLOK Dataset.

C. Discussion

The results from both the eLOK and ASAP-SAS datasets demonstrate that the Atomic Evaluation method effectively improves the Quadratic Weighted Kappa (QWK) scores for both Llama 3 8B and Gemma 2 9B models. This shows that the method enhances the accuracy and consistency of small parameter LLMs in Automated Essay Scoring (AES) tasks.

However, despite its demonstrated effectiveness, the Atomic Evaluation method introduces longer processing times due to the multiple steps involved in decomposing both questions and answers into smaller units. This can be a limitation, particularly in large-scale applications or scenarios requiring real-time scoring, where efficiency is critical. Therefore, it is recommended that future research explore ways to reduce the processing time of the Atomic Evaluation method to make it more suitable for such settings. One potential solution could be to explore the use of models with smaller parameters, as they may offer improved efficiency without significantly compromising accuracy. Additionally, optimizing the atomic decomposition process itself, by streamlining the extraction and evaluation of atomic units, could further reduce the computational load. More efficient prompt-engineering techniques may also reduce the interaction steps required between the model and the dataset, thereby speeding up the overall process.

The improvement in QWK scores is significant primarily for models with relatively small parameters, such as Llama 3 8B and Gemma 2 9B. The Atomic Evaluation method aids these smaller models in overcoming their limitations in handling longer contexts by breaking down complex answers into manageable atomic units. However, for larger parameter models that already perform well with longer contexts, the benefits of Atomic Evaluation may be less pronounced. Thus, future research should focus on optimization strategies tailored to both smaller and larger models to maximize their performance in different contexts.

To further validate the effectiveness of Atomic Evaluation, future work could explore its application across a broader range of small parameter LLMs and datasets from various domains, such as legal, medical, or technical education. This would help assess the method's versatility and scalability. By optimizing prompt techniques and exploring the use of smaller models, Atomic Evaluation could provide a more efficient, scalable solution for educational environments with diverse computational constraints.

V. CONCLUSION

This study introduced a novel approach called Atomic Evaluation to improve the performance of small parameter Large Language Models (LLMs) in automated essay scoring tasks. The Atomic Evaluation method enhances accuracy by decomposing both questions and student responses into smaller, more manageable atomic units, enabling more granular and focused evaluations.

The primary contribution of this work is demonstrating that Atomic Evaluation consistently improves the Quadratic

Weighted Kappa (QWK) score of small parameter LLMs compared to the traditional Chain of Thought (CoT) method. Specifically, the QWK scores improved by an average of 0.0548 for Llama 3 8B and 0.2009 for Gemma 2 9B on the eLOK dataset, and by 0.0437 and 0.0299 on the ASAP-SAS dataset, respectively.

The implications of this research are significant for educational environments, particularly those with limited computational resources. Atomic Evaluation provides a cost-effective, efficient solution, allowing smaller models to achieve greater accuracy in handling complex essay scoring tasks. This reduces reliance on large-scale models, lowers computational costs, and ensures reliable, scalable performance, making it highly suitable for widespread adoption in resource-constrained educational settings.

Future directions for this research include further optimizing the Atomic Evaluation process to reduce computational time and enhance efficiency, particularly in real-time applications. Additionally, expanding the application of Atomic Evaluation to a broader range of datasets and domains, such as legal, medical, or technical education, could help assess its adaptability and scalability across various contexts.

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